COFFS HARBOUR, Australia

WMO: **947910**

Lat: **30.311S** Lon: **153.119E** Elev: **6** StdP: **101.25** Time Zone: **10.00 (AUS)** Period: **95-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	7	4.1	5.5	-2.0	3.2	14.7	-0.1	3.8	14.0	11.0	16.8	10.0	16.2	1.8	310	0.433	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	1	6.8	30.1	22.2	28.6	22.2	27.7	22.0	24.3	27.7	23.7	26.9	23.1	26.3	6.5	40	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	23.2	17.9	26.3	22.6	17.3	25.8	22.0	16.7	25.3	73.6	27.8	70.9	27.1	68.6	26.3	31.0	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	11.1	9.9	8.9	DB	1.6	35.8	0.9	2.6	1.0	37.6	0.4	39.2	-0.1	40.6	-0.7	42.5	(4)
(5)				WB	1.2	25.9	0.9	1.5	0.5	27.0	0.0	27.9	-0.5	28.7	-1.2	29.8	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	18.8	23.4	23.2	22.1	19.5	16.6	14.5	13.5	14.3	16.8	18.9	20.6	22.2	(6)
(7)			DBStd	4.08	2.03	1.84	1.69	1.84	2.23	2.20	2.23	2.39	2.48	2.43	2.40	2.18	(7)
(8)			HDD10.0	2	0	0	0	0	0	1	1	1	0	0	0	0	(8)
(9)			HDD18.3	551	0	0	0	8	62	116	152	127	59	22	5	1	(9)
(10)			CDD10.0	3204	414	369	376	287	204	136	108	134	205	275	318	378	(10)
(11)			CDD18.3	712	156	136	118	44	8	1	1	3	14	39	73	121	(11)
(12)			CDH23.3	3479	924	766	513	133	20	2	4	17	73	155	288	584	(12)
(13)			CDH26.7	499	142	112	52	7	1	0	0	3	15	30	54	84	(13)
(14)	Wind		WSAvg	4.2	4.8	4.5	4.2	3.9	3.5	3.7	3.6	3.9	4.2	4.7	4.9	4.8	(14)
(15)	Precipitation		PrecAvg	1671	179	204	235	187	155	125	69	69	56	103	153	141	(15)
(16)			PrecMax	2882	513	522	782	752	535	530	375	306	226	495	547	395	(16)
(17)			PrecMin	1071	12	51	20	15	10	8	0	0	0	8	24	33	(17)
(18)			PrecStd	437	118	116	147	158	127	108	73	73	50	96	95	91	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		32.0	31.4	30.0	27.3	25.2	23.2	23.5	25.4	29.0	30.1	32.1	31.5	(19)
(20)			MCWB		23.2	24.1	23.1	20.6	18.3	16.2	15.8	15.8	17.6	18.4	19.3	22.3	(20)
(21)		2%	DB		29.1	29.1	28.0	26.0	23.6	21.3	21.1	22.7	25.2	26.7	27.9	28.6	(21)
(22)			MCWB		23.2	23.3	22.7	20.1	17.6	15.8	14.6	15.0	17.2	18.9	20.7	22.0	(22)
(23)		5%	DB		28.0	27.9	26.9	24.9	22.2	20.1	19.8	21.1	23.3	24.8	26.0	27.1	(23)
(24)			MCWB		22.9	22.8	22.1	19.5	17.0	15.1	14.0	14.6	17.0	18.8	20.7	21.8	(24)
(25)		10%	DB		27.1	27.0	25.9	23.9	21.2	19.2	18.6	19.9	22.0	23.4	24.8	26.1	(25)
(26)			MCWB		22.3	22.1	21.4	18.9	16.4	14.4	13.4	14.1	16.4	18.2	20.1	21.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		25.4	25.1	24.5	22.1	20.2	18.2	17.7	18.3	20.2	21.7	22.8	24.5	(27)
(28)			MCDB		28.5	28.9	28.1	25.7	22.9	20.9	20.4	22.3	24.4	25.3	26.8	28.3	(28)
(29)		2%	WB		24.3	24.2	23.5	21.1	19.2	17.1	16.5	16.9	19.1	20.5	22.0	23.4	(29)
(30)			MCDB		27.5	27.7	26.6	24.5	22.0	19.8	19.3	20.4	22.7	24.4	25.9	26.9	(30)
(31)		5%	WB		23.6	23.6	22.7	20.4	18.3	16.3	15.4	15.8	18.2	19.7	21.2	22.6	(31)
(32)			MCDB		26.8	26.9	25.8	23.7	21.2	18.9	18.4	19.5	21.8	23.3	24.7	25.9	(32)
(33)		10%	WB		23.0	22.9	22.0	19.6	17.3	15.5	14.3	15.0	17.4	19.0	20.6	22.0	(33)
(34)			MCDB		26.1	26.0	25.1	22.8	20.0	18.2	17.4	18.7	21.0	22.4	23.9	25.1	(34)
(35)	Mean Daily Temperature Range		MDBR		6.8	6.7	7.2	8.1	9.2	9.3	10.5	11.1	10.2	8.7	7.5	7.3	(35)
(36)		5% DB	MCDBR		8.1	8.1	8.4	9.6	10.3	10.2	12.0	12.9	12.2	11.2	9.4	8.7	(36)
(37)			MCWBR		4.0	3.8	4.3	5.0	5.5	5.8	6.8	7.0	6.5	5.7	4.7	4.5	(37)
(38)		5% WB	MCDBR		7.1	7.2	7.4	8.4	8.3	8.3	9.5	10.8	10.2	9.5	8.1	7.7	(38)
(39)			MCWBR		4.0	3.9	4.2	4.7	5.0	5.3	6.0	6.5	6.0	5.6	4.5	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.394	0.379	0.360	0.338	0.317	0.308	0.302	0.314	0.358	0.393	0.389	0.393	(40)
(41)		taud			2.491	2.550	2.596	2.631	2.638	2.656	2.653	2.602	2.475	2.407	2.477	2.484	(41)
(42)		Ebn at Noon			947	943	929	901	875	862	884	913	915	918	946	951	(42)
(43)		Edh at Noon			116	106	95	83	74	69	72	84	106	121	117	117	(43)
(44)	All-Sky Solar Radiation	RadAvg			6.55	5.90	4.96	4.17	3.37	2.80	3.20	4.12	5.24	5.87	6.27	6.53	(44)
(45)		RadStd			0.50	0.47	0.34	0.25	0.23	0.18	0.26	0.27	0.34	0.51	0.56	0.56	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+141	+86	(46)
(47)	Regional (13 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-61	+181	+92	(47)

Nomenclature: See separate page